

Hassan Raza

MACHINE LEARNING ENGINEER · DATA SCIENTIST

Lahore · Pakistan

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Summary

Data Scientist and Machine Learning Engineer with over 4 years of experience in developing data-driven solutions for global clients, specializing in machine learning (ML), deep learning (DL), and data analysis. Proficient in processing and preparing large-scale datasets, building predictive models, and delivering actionable insights. Adept at collaborating with cross-functional teams to design dashboards and prototypes that bridge business gaps, enhance decision-making, and drive growth. Skilled in managing diverse datasets for international industries and markets.

Work Experience

AiSight.ai

DATA SCIENTIST

SaaS-based Startup

Jan. 2023 - Present

- **RTM Analytics Platform**

Architected and deployed a multi-tenant internal web platform (FastAPI + HTMX, Docker, AWS S3) serving field operations teams across FMCG markets worldwide. Delivered 20+ geospatial and data processing tools, replacing manual analyst workflows with a self-serve UI. Implemented async job execution, SQLite WAL-mode persistence, and S3 integration.

- **Business Insights for Nielsen and UK Clients**

Collaborated with Nielsen and UK clients to develop prototypes, extract actionable insights, and improve business strategies. Implemented a chatbot using NLP and LLM modeling for automated data queries. Led a team of 2-3 engineers and delivered the project on time.

- **Data Scraping and AI-Powered Sales & Footfall Analysis**

Scraped and analyzed shop-level data from multiple web sources using Python, including footfall statistics, building area, morning/evening population density, and peak rush times. Integrated these features with AI-driven models to generate explainable business insights, guiding strategic sales planning and improving regional competitiveness.

- **Turning EBM Sales Data into Strategic Business Insights**

Collaborated with cross-functional teams at EBM to design interactive dashboards, offering valuable sales insights. Used deep learning models to detect sales patterns and provide actionable suggestions for market growth. Additionally, applied satellite imagery and computer vision to identify potential market areas, and delivered alerts for optimizing demand forecasting and improving sales performance.

- **ETL Automation & Scalable Data Pipelines**

Led the implementation of automated data extraction, transformation, and loading (ETL) processes utilizing Apache Airflow. Designed and executed scalable, fault-tolerant data pipelines, integrating diverse data sources and optimizing processing speed through parallel computing techniques to enhance overall data integrity and efficiency.

SurveyAuto Inc

MACHINE LEARNING ENGINEER

SaaS-based Startup

Jan. 2021 - Dec 2022

- **Statistical Sales Forecasting**

Statistically forecasted sales across multiple views of the data; for example, at the product level, brand level, and company level, as well as against sales within a specific geospatial area.

- **Semi Supervised Territory Definition for Optimal Distribution**

Shop coordinates are clustered under a time constraint so that distributors can visit the maximum number of desired shops for distribution within a given time frame.

- **Satellite Image Analysis for Population Estimation**

Utilized satellite imagery to extract area images and applied machine learning techniques to identify and estimate population density at a granular level within specific areas.

Code District

SOFTWARE ENGINEER

Software Service Provider

July. 2020 - Dec.2020

- **Revitalize Charging Solutions**

Developed a backend server for a US-based company managing multiple revenue-generating websites in the electric vehicle (EV) space. The system enabled users to locate and pay for EV charging services through a centralized platform. Integrated secure payment processing and API communication to support seamless real-time charging requests and transactions.

Education

Punjab University College of Information Technology (PUCIT)

Lahore, Pakistan

MS IN COMPUTER SCIENCE

Oct. 2020 - Aug. 2022

- **Automatic Speech to Text Using Natural Language Processing & Deep Learning** [THESIS]

I trained multiple variations of recurrent models on an Urdu speech dataset to generate the text of the speech from the speech spectrums. The following NLP & deep learning techniques were used in this work: Bi-LSTMS, Local & Global Attention Mechanisms, Copy Mechanism, Beam Search, and Natural Text Preprocessing.

- **Courses**

Machine Learning, Deep Learning, Big Data, Statistics & Probability, Natural Language Processing, Computational Modeling

Punjab University College of Information Technology (PUCIT)

Lahore, Pakistan

BS IN COMPUTER SCIENCE

Sep. 2016 - Jun. 2020

- **Optical Character Recognition for URDU** [PROJECT]

Designed and implemented a research-driven Optical Character Recognition (OCR) system for Urdu script. The project consisted of two modules: (1) Data collection and preprocessing - captured Urdu character images from real-world sources to create a custom dataset, and (2) Model development — built a computer vision pipeline to recognize Urdu characters in real-world scenarios using classical image processing and pattern recognition techniques.

- **Courses**

Computer Programming, Object Orientations, Data Structures, Algorithms, Databases, Networking

Skills & Tools

RECENTLY & FREQUENTLY

- **Skills:** Software Development, Object Oriented Programming, Machine Learning, Explainable Statistical Modeling, Data Analysis, Data Engineering, Data Pipelining, Data Visualization, Semantic Code Refactoring, Runtime Performance Tuning
- **Tools:** Python3, Pandas, Jupyter Lab, IPython, Numpy, Matplotlib, Tensorflow, Scikit-learn, Linux, VS Code

OCCASIONALLY

- **Skills:** Deep Learning, Natural Language Processing, Geo-Spatial Analysis, Databases, Designing Systems from Scratch, Products Deployment, Automation, Web Scraping
- **Tools:** Pytorch, SciPy, Docker, Apache Airflow, MySQL, MongoDB, Scrapy, Javascript, HTML, CSS

Certifications

Machine Learning

AndrewNG, Coursera, Stanford

Online

SUPERVISED LEARNING TECHNIQUES

July. 2021

- Linear and non-linear models for supervised classification and regression.
- Optimization techniques and tweaks for better generalization.